Gunj: dating Indian recordings from mains hum against an open grid-frequency record

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Code, protocol and results: https://github.com/satanrayshe/gunj · Explorer: https://satanrayshe.github.io/gunj/
Entry to Threws Odyssey 2026, Cybersecurity track. Test results as of the scoring run of 11 Nov 2026, 19:16 IST.

Abstract. Mains hum in a recording follows the frequency of the power grid, and India runs one synchronous grid. Gunj extracts a 1-Hz trace of the electric network frequency (ENF) from audio and searches a public one-second log of Indian grid frequency (20 Dec 2020 to 20 Aug 2023) for the start time that fits. The log's own time labels drift late by up to 319 s, and Gunj corrects each recording with a drift estimate measured against the grid's 15-minute scheduling clock. In a pre-registered, blind evaluation on public livestreams of Indian institutions, 126 of 150 selected Test streams were harvested before a bot check stopped the downloads. For 60-min excerpts searched over the whole log, the top candidate started within 30 s of the expected time for 25 of 90 streams (27.8%, 95% CI 19.6 to 37.8%), and for 5 of 90 without the drift correction. Gunj gave a confident verdict on 23 of these excerpts, 14 of them within 30 s and 9 further off, and 1 of 16 real foreign 50 Hz recordings reached its threshold.

Keywords: electric network frequency · audio forensics · timestamp verification · pre-registration · India

1 Introduction

Recordings of public events are often shared again long after they were made, and the metadata that could show when a recording was made can be removed or edited. Mains hum offers a physical timestamp: the frequency of an AC grid wanders around its nominal value as load and generation change (standard deviation 73.6 mHz around 50 Hz in the log used here), a recorder near mains wiring picks up hum at that frequency, and on one synchronous grid the wander is the same everywhere. Matching the frequency trace of a recording against a log of grid frequency can therefore date the recording, which is the ENF criterion of forensic audio analysis [1].

India runs one synchronous grid, and a one-second log of its frequency is published under an open licence [2]. This paper describes Gunj, an open implementation of ENF dating against that log, with four contributions: (i) a reference series built from the published files, with its gaps mapped; (ii) the finding that the log's time labels drift by minutes, with a per-recording correction measured against the grid's 15-minute scheduling clock (Sections 3 and 4); (iii) a blind evaluation on public livestreams of Indian institutions, fixed before any Test audio existed and reporting every selected stream (Sections 5 and 6); (iv) a record of every change to that plan (Section 7). Gunj says “no verdict” when a recording has no hum or its confidence is below a threshold fixed in advance, and it reports time and grid only.

2 Related work

Grigoras [1] introduced the ENF criterion for authenticating and dating digital audio against a database of grid frequency. Huijbregtse and Geradts [3] studied the dating of short recordings with correlation-coefficient matching, and Rodríguez, Apolinário and Biscainho [4] located edits from discontinuities in the ENF phase. Hajj-Ahmad, Garg and Wu [5] combined the estimates of several harmonics, each weighted by its local signal-to-noise ratio, as Gunj's v2 extractor does frame by frame. Garg et al. [6] recovered ENF from video through the flicker of mains lighting. Hajj-Ahmad et al. [7] identified the grid of a recording without a concurrent reference, a location-of-recording task that the IEEE Signal Processing Cup 2016 also set [8], and Kim, Jeon and Yoon [9] built a national-scale ENF map from online multimedia streams. Open tools such as libhum [10] and hum-catcher [11] match a trace against a grid-frequency reference that the user supplies. In a search on 5 Nov 2026 we did not find an evaluation of ENF dating against an Indian grid log. Gunj adds an open Indian reference that can be rebuilt from the published files, the label-drift correction, and a pre-registered blind evaluation with a “no verdict” outcome; it has not been compared with another ENF tool, because the planned libhum baseline was not run (Section 7).

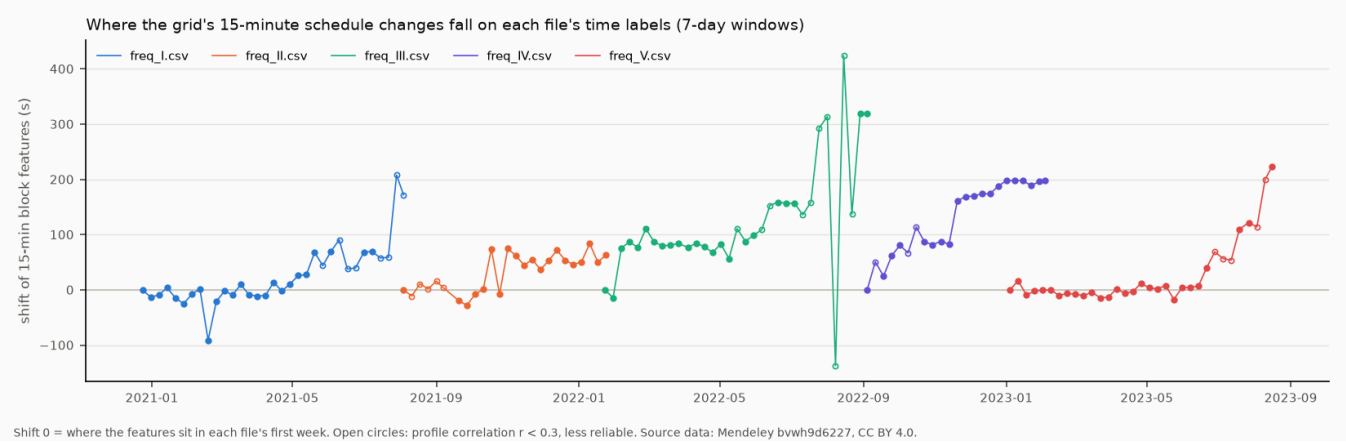


Fig. 1. Label drift of the reference against the grid's 15-minute scheduling clock, in weekly windows. Each series is one source file, zero at its first week; open circles mark windows with a weak profile correlation. At the end of each file the labels run late by 172, 64, 319, 198 and 224 s (files I to V). Source: results/time_axis_check.json.

3 Data

3.1 Reference and its label drift

The reference is Mendeley Data dataset bvwh9d6227 [2]: one-second frequency from a phasor measurement unit (PMU) of Grid-India's North-Eastern Regional Load Dispatch Centre, shared by IIT Guwahati under CC BY 4.0. Its 5 CSV files were merged into one series that runs from 2020-12-20 17:21:01 to 2023-08-20 11:33:05 IST: 84,046,325 seconds (972.76 days), of which 99.08% are valid. The series has 61 gaps longer than 10 s, holding 772,056 missing seconds; the longest lasts 420,458 s, from 2021-09-11 13:22:38 to 2021-09-16 10:10:15 IST.

The time labels of the dataset are not consistent. Where consecutive files overlap, they hold the same values under labels 172, 64, 319 and 198 s apart, and inside each file the labels drift late in steps. We measured the drift against the grid's 15-minute scheduling clock: frequency changes faster just after each block boundary, so folding the absolute second-to-second change on time mod 900 s gives a profile whose peak moves when the labels drift (Fig. 1). Relative to its first week, which the check assumes to be on time, each file runs late by 172, 64, 319, 198 and 224 s at its end (files I to V), and at the overlaps the clock check gives the same four offsets within 1 s.

3.2 Streams, splits and exclusions

Candidate streams were found from public YouTube metadata alone, before any audio existed: 424 Indian livestreams from 31 channels and 116 foreign livestreams from 6 channels on 50 Hz grids. Court channels and proceedings were excluded (Section 8), as were streams of 11.5 h or longer, which may have hit YouTube's archive length cap (23 removed), and any stream for which less than 90% of the reference seconds over its whole span are valid (5 removed). One seeded draw (seed 2026110701) from the 396 eligible Indian streams of 30 channels formed a Dev split of 5 streams for tuning, a Calibration split of 20 for the latency and a Test split of 150 streams from 28 channels, at most 10 per channel and none overlapping in time: 57 legislature, 37 lecture, 30 government event, 24 religious and 2 conference streams. The negatives are 50 foreign 50 Hz livestreams from the United Kingdom (16), Italy (11), Sri Lanka (10), Pakistan (10) and Vatican City (3), assigned alternately to a threshold half and a report half, and 40 SYNTHETIC items (20 of SYNTHETIC white noise and 20 of SYNTHETIC hum that follows a SYNTHETIC random walk), also halved. Only the first 65 min of audio were downloaded, without sign-in or cookies, converted to 1 kHz mono, and, apart from the Dev audio, deleted after the traces were extracted. Of the 150 Test streams, 126 have a trace; of the other 24, 10 were no longer available on YouTube, 8 failed in download or conversion, 5 had not been attempted and 1 met the bot check that stopped the harvest.

4 Method

4.1 Extraction

Extractor v1 uses 16 s Hann windows with a 1 s hop and a 2^{18}-point FFT at 1 kHz. In each frame it takes a peak for each of harmonics 1 to 6 within k·50 ± 0.15k Hz, refines it by quadratic interpolation and divides it by k; the harmonics are combined with weights max(median peak-to-floor − 2, 0), and a 5-point running median smooths the trace. The hum flag of a recording is set if any harmonic's median peak-to-floor ratio ≥ 3. Extractor v2 drops the harmonics whose median ratio is below 3 and weights the others in each frame by $\max(\mathrm{SNR} - 1, 0)^2$, where SNR is the peak-to-floor ratio in that frame; the total weight of a frame, smoothed and normalised, also weights the correlation. v2 was chosen from 10 candidate settings on the Dev streams only, by the mean clipped Fisher-z margin J of the true start over the best other candidate (0.222, against −0.012 for v1 and −0.046 for a naive single-bin 50 Hz extractor), and it was frozen with a git tag before any Calibration, Test or negative audio was extracted, so it is the primary extractor.

4.2 Matching, confidence and verdict

The trace is compared with every position in a search scope by normalised cross-correlation computed with FFTs (weighted for v2). Windows with less than 90% valid reference seconds or a reference standard deviation under 0.003 Hz are skipped, and the 5 best local peaks are kept as candidates. The confidence is the Fisher-z margin $c = \operatorname{atanh}(r_1) - \operatorname{atanh}(r_2)$ between the top candidate and the best candidate at least 60 s from the top candidate. A raw correlation is a poor score on a log this long: the best real threshold-half negative reached a top correlation of 0.9951 (v2, 10 min, full span) with a confidence of only 0.0022. The verdict is “match” if the hum flag is set and $c \geq \tau$, where τ depends on the extractor, excerpt length and scope (Section 5), and “no verdict” otherwise. Excerpt lengths are 10, 20, 40 and 60 min; an m-minute excerpt covers audio seconds [120, 120 + 60m), is extracted on its own, and is searched over the full span of the log, over ±90 days and over ±1 day around the release time in IST, the last being a check of a claimed date.

4.3 Ground truth, label drift D and latency L

The expected start of a recording on the log's labels is T* = release time (IST) + L + D, and the error e of a candidate is its implied start minus T*. A hit is $|e| \leq 30$ s; $|e| \leq 120$ s and $|e| \leq 600$ s are reported as coarse classes and never counted as hits. D is the label drift at the stream's start, read from the clock check of the file in use after a running median over 3 windows and frozen in the split file before any audio existed; a stream is label-stable if the profile correlation of that window is at least 0.5. For 93 of the 150 Test streams |D| exceeds 30 s, more than the tolerance of a hit ($|e| \leq 30$ s; median |D| 58 s, largest 319 s), and 39 are label-stable. L is one constant delay between YouTube's release time and the first audio sample, estimated on the Calibration split only as the median offset o of the streams that lock with $c \geq \tau$ and $|o| \leq 600$ s. It came out as L = 2 s, from 5 of the 19 harvested Calibration streams, whose offsets run from −4 s to 5 s (interquartile range 0 s to 2 s).

4.4 Edits and software

For edit detection (M5, tested on SYNTHETIC cuts only), find_edits splits a locked trace into pieces that fit the reference at offsets 5 s or more apart; a positive jump means missing audio, that is, a cut. Split points are searched at least 180 s from either end of the trace, and refining a split can move it up to 26 s closer, so a cut within 154 s of either end cannot be found. The code uses NumPy [12], SciPy [13], pandas [14] and Matplotlib [15]; audio was fetched with yt-dlp [16] and converted with FFmpeg [17].

5 Evaluation protocol

The plan, PROTOCOL.md in the repository, was approved and committed on 7 November 2026 at 09:05 IST (commit 2485706), before any Test or negative audio existed. It fixes the splits by their SHA-256 hash, the seeds, the metrics, the hit rule, the threshold rule and a go or no-go gate, and PROTOCOL section 19 records every later change to them (Section 7).

Thresholds. τ is set for each extractor, excerpt length and scope on the threshold half of the negatives. With n trials in a cell, m = floor(0.01 n) and τ is the (m+1)-th largest confidence plus 0.0001, so that at most 1% of the threshold-half trials reach it; with n under 100, m = 0, so τ sits just above the largest negative confidence of the cell. τ was set on 23 real and 20 SYNTHETIC threshold-half negatives (2 real ones could not be downloaded). In the primary cell τ = 0.0834.

Blinding. τ, L and the scoring script were committed before any Test search ran (commits beadde1, 84ac47d and 6d010bb), and the script checks with git that they are unchanged before it reads a Test trace. Test streams were harvested in a seeded random order, so the streams harvested by any time form a random sample of the split.

Metrics. M1 is hum presence; M2 the top-1 hit rate by excerpt length and scope, with the primary endpoint M2 for v2, 60-min excerpts and the full scope; M3 the false-match rate of the report-half negatives at τ (target at most 1%), with a reliability plot; M4 the excerpt length needed for 90% unique locks, simulated with SYNTHETIC noise on real reference excerpts and read at the noise measured on the Test streams that lock; M5 edit detection on SYNTHETIC cuts of 10 to 120 s; and M6 an optional live station. Every rate has a Wilson 95% interval [18]; no significance test is run. Sensitivity analyses: S1 drops D, S2 keeps label-stable streams only, S3 drops the 27 Test streams of channels whose official status or location is uncertain, and S4 compares v1 with v2. Baselines: v1, a naive single-bin 50 Hz extractor with its own τ, libhum, and an analytic random guess.

Gate. Work would continue only if at least 5 independent Test streams locked blind (a top-1 hit with $c \geq \tau$ in the full or ±90-day scope, at any excerpt length) and harvesting met no bot check, CAPTCHA, sign-in wall or HTTP 429.

6 Results

6.1 Gate and hum presence (M1)

All Test numbers are final, since no further download is planned, and come from the scoring run of 11 Nov 2026, 19:16 IST (results/evaluation.json and results/test_results.csv). The frozen gate run (8 Nov 2026, 17:04 IST) found 38 of 126 Test streams locking blind (9 with D set to 0), against 19 for v1 and 13 for the naive extractor, so the lock conditions were met. The harvest-health condition failed: YouTube answered 1 stream with a bot check on 8 Nov 2026, 05:30 IST, and the harvester stopped. The gate was therefore not met. The plan then called for a switch to a fallback project; Section 7 records the decision to continue. The M1 hum flag was set for 50 of 126 Test streams with audio (39.7%, 95% CI 31.6 to 48.4%); by venue type, legislature 28 of 52, lecture 5 of 29, government event 6 of 25, religious 11 of 19 and conference 0 of 1. As description only, the flag was set for 19 of 47 harvested real foreign negatives (40.4%, 95% CI 27.6 to 54.7%).

6.2 Dating accuracy (M2)

Table 1 gives M2 for every cell of v2. The primary endpoint is 25 of 90 streams (27.8%, 95% CI 19.6 to 37.8%), the same over ±90 days; among streams with the M1 hum flag it is 16 of 37 (43.2%, 95% CI 28.7 to 59.1%). Of the 90 excerpts, the top candidate is within 30 s for 25, 30 to 120 s off for 7, 120 to 600 s off for 2 and further off for 55 (54 by more than a day); 1 excerpt has no candidate, because the trace was constant. Gunj gave a confident verdict for 23 excerpts and no verdict for 67; of the 23, 14 are within 30 s, 6 are 30 to 120 s off, 1 is 120 to 600 s off and 2 are off by 5,299 s (88 minutes) and 2,876,519 s (33.3 days). Confident hits are thus 14 of 90 (15.6%, 95% CI 9.5 to 24.4%) and confident wrong verdicts 9 of 90 (10.0%, 95% CI 5.4 to 17.9%).

Table 1. M2 of extractor v2: top-1 candidates within 30 s of T*, as hits, share of N and Wilson 95% interval. N counts the Test streams with a trace at that length; the primary cell is in bold. The last three columns split the confident verdicts (match) of the full span into hits (within 30 s) and wrong verdicts. Sources: results/evaluation.json, results/error_breakdown.json.

		Top-1 within 30 s of T*: hits · share [Wilson 95%]			Confident, full span		
Excerpt	N	Full span	±90 days	±1 day	Match	Hit	Wrong
10 min	126	28 · 22.2% [15.8, 30.2]	28 · 22.2% [15.8, 30.2]	35 · 27.8% [20.7, 36.2]	19	17	2
20 min	126	31 · 24.6% [17.9, 32.8]	31 · 24.6% [17.9, 32.8]	40 · 31.7% [24.3, 40.3]	21	19	2
40 min	104	26 · 25.0% [17.7, 34.1]	26 · 25.0% [17.7, 34.1]	30 · 28.8% [21.0, 38.2]	24	16	8
60 min	90	**25 · 27.8% [19.6, 37.8]**	25 · 27.8% [19.6, 37.8]	28 · 31.1% [22.5, 41.3]	23	14	9

6.3 Baselines and sensitivity analyses

Table 2 compares v2 with the baselines on the same 90 streams and gives the sensitivity analyses. Every stream that v1 (15 hits) or the naive extractor (12) puts within 30 s is also a hit of v2. A uniform random guess would hit with probability 7.258×10^{-7} per stream, or 6.5×10^{-5} times over these 90 streams. Without the drift correction (S1) the endpoint falls to 5 of 90 (5.6%, 95% CI 2.4 to 12.4%).

Table 2. Baselines and pre-registered sensitivity analyses in the primary cell (60-min excerpts, full span). The paired column compares each baseline with v2 on the same streams. Sources: results/evaluation.json, results/test_results.csv, results/splits_summary.json.

Method or analysis	Top-1 within 30 s: hits/N · share [95% CI]	Confident hit	Confident wrong	Paired with v2: both / v2 only / other only / neither
Extractors on the same streams				
v2, primary	25/90 · 27.8% [19.6, 37.8]	14 (15.6%)	9 (10.0%)	
v1, frozen extractor	15/90 · 16.7% [10.4, 25.7]	11 (12.2%)	4 (4.4%)	15 / 10 / 0 / 65
naive single-bin 50 Hz	12/90 · 13.3% [7.8, 21.9]	8 (8.9%)	2 (2.2%)	12 / 13 / 0 / 65
libhum	not run: libhum was never installed, and the time went to the harvest problems			
random guess, analytic	$p = 7.258 \times 10^{-7}$ per stream; 6.5×10^{-5} expected hits over these 90 streams			
Sensitivity analyses, extractor v2				
S1: no drift correction (D = 0)	5/90 · 5.6% [2.4, 12.4]			
S2: label-stable streams only	9/25 · 36.0% [20.2, 55.5]			
S3: verified channels only	22/75 · 29.3% [20.2, 40.4]			
S4: v1 against v2	the paired counts above			

6.4 False matches (M3)

The report half was searched with 24 of its 25 real foreign streams and all 20 SYNTHETIC items. In the primary cell τ was reached by 1 of 16 real negatives (6.2%, 95% CI 1.1 to 28.3%), 0 of 20 SYNTHETIC ones (0.0%, 95% CI 0.0 to 16.1%) and 1 of 36 together, so the pre-registered target of at most 1% is not met. Across the 12 cells of v2 a real negative reached τ in 4 cells and a SYNTHETIC one in 3; adding the hum-flag condition changes no count. No cell has enough negatives to show a rate below 1%: the smallest Wilson upper bound, real and SYNTHETIC together, is 8.0%. The confidence still separates Test hits from misses (Fig. 2). Test excerpts with c below 0.025 hit in 0 of 46 cases and those with c of 0.1 or more in 22 of 30. In the primary cell the largest confidence of a real negative was 0.1339 and of a SYNTHETIC one 0.0394, against τ = 0.0834.

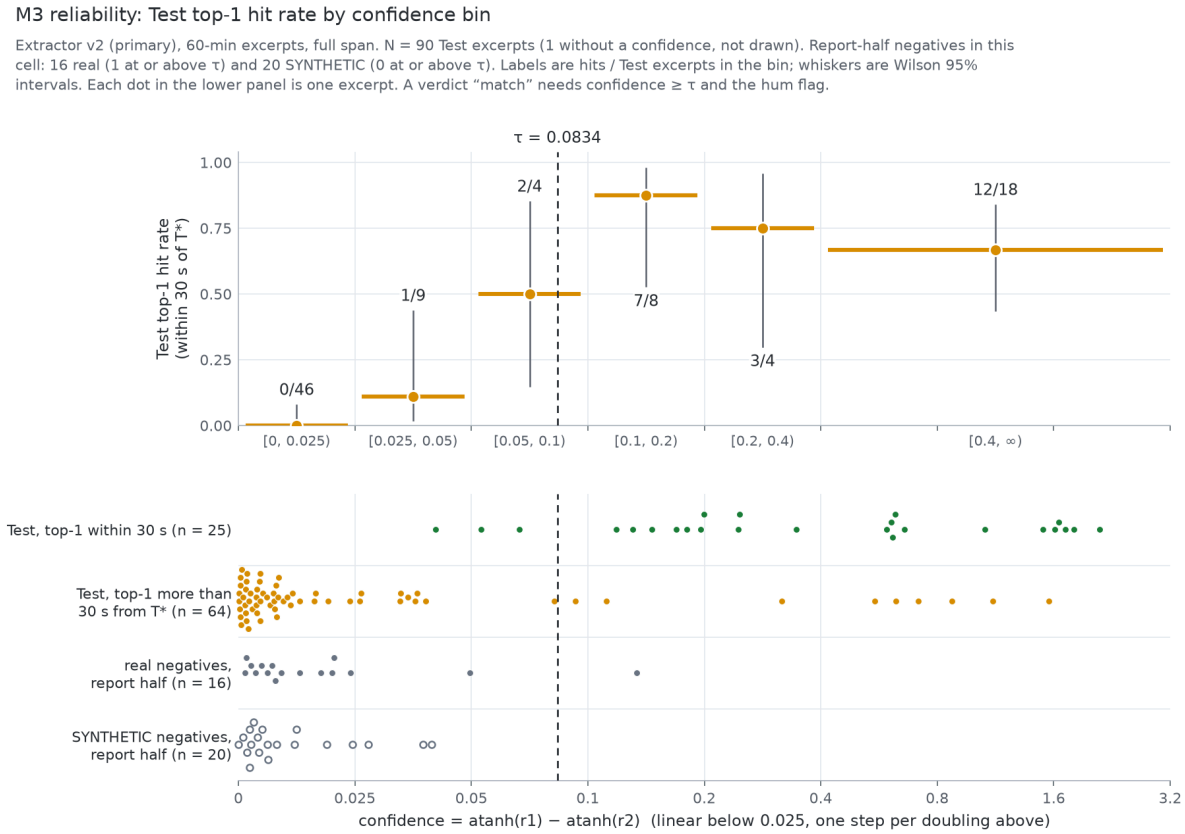


Fig. 2. M3 reliability plot for the primary cell. Top: Test top-1 hit rate by pre-registered confidence bin, with Wilson 95% intervals. Bottom: the confidence of every Test excerpt and of the report-half negatives, 16 real (filled) and 20 SYNTHETIC (open). Dashed line: τ = 0.0834, with 1 real negative to its right. Source: results/figures/figures.json.

6.5 Audio needed for a unique lock (M4)

Before any Test search the simulation was read at 40 mHz, the tested level nearest the 38.6 mHz equivalent trace noise of the feasibility probe; with SYNTHETIC noise smoothed over 16 s, 90% unique locks over the full span needed 40 min of audio (422 of 450 trials, 93.8%). The plan then reads the simulation at the noise measured on the Test streams that lock, computed as in the script that analysed the feasibility probe, which uses the plain Pearson correlation of a trace with its matched window. Read that way (reading B), the median noise over the 38 streams (88 locking excerpts) is 37.4 mHz (quartiles 12.9 to 71.2), and the shortest tested length with at least 90% unique locks over the full span is 40 min (94.6%;

20 min gives 35.7%). EXPLORATORY (post hoc): with v2's weighted correlation in the same formula (reading A), the median is 9.4 mHz and the length 10 min (91.2%; 5 min gives 13.3%). This noise comes from streams that lock, so it describes recordings with usable hum. The M2 and M3 plots of every cell and the M4 plots are in results/figures and results/uniqueness_curve.png.

6.6 Edit detection on SYNTHETIC cuts (M5) and other results

Every cut is SYNTHETIC, a stretch of 10 to 120 s removed from the 60-min excerpt audio at harvest time. On the 14 Test streams whose uncut 60-min excerpt is a confident hit over the full span, 14 of 28 SYNTHETIC cuts (50.0%, 95% CI 32.6 to 67.4%) were detected within 30 s of their true position; none of the 3 SYNTHETIC cuts of 10 s was detected. For the detected SYNTHETIC cuts the missing duration was off by a median of 1.5 s and at most 5 s. As M5 false alarms, find_edits reported an edit on 2 of 14 uncut excerpts (14.3%, 95% CI 4.0 to 39.9%) of the same streams. No recording with a real edit was tested. M6, the optional live station, was not run.

EXPLORATORY (post hoc). In the primary cell the drift corrections |D| of the 14 confident verdicts within 30 s (median 60.5 s, range 0 s to 293 s) and of the 7 that are 30 to 600 s off (median 67 s, range 8 s to 200 s) overlap; these are counts with no test, and they cannot show whether an offset comes from YouTube's start time, L, D or the lock. Over all 31 gate locks with SYNTHETIC cut traces, M5 detects 16 of 62 SYNTHETIC cuts.

7 Deviations and threats to validity

PROTOCOL section 19 records 10 changes made after the plan, each with its time (IST), its reason and whether any Test result had been seen.

1. **7 Nov 2026, 08:30 IST, before approval.** Dev and Calibration audio were harvested before the plan was approved, under the author's approval of audio downloads given at 01:20 IST, and v2 was frozen at 05:55 IST; no Test or negative audio existed. Calibration traces were searched before τ existed, but L was set only after τ was committed; the local history was rewritten before any push to remove local paths and edit commit messages, with no code change, and the plan gained a fixed harvest order.
2. **7 Nov 2026, 10:00 IST.** The split file is sorted by date, so its row order would have harvested the oldest Test streams first. The Test harvest used a seeded order instead (SHA-256 of the split seed and the video ID), so the streams finished by the gate are a random sample.
3. **7 Nov 2026, 11:50 IST.** The laptop lost its internet connection from 10:18 to 11:38 IST, and a τ written during the outage from 1 real and 20 SYNTHETIC threshold-half negatives was deleted before it was committed or used. Network loss no longer counts as a failed download.
4. **7 Nov 2026, 12:15 IST.** A read-only code review led to harvester fixes (rate-limit messages count as a block, private videos as unavailable, a download with a skipped segment as failed, and a stop after 4 real downloads in a row fail), and the attempts charged during the outage were given back to 182 rows of the manifest. The rules for τ, L, hits and the gate did not change.
5. **7 Nov 2026, 12:30 IST.** The same review led to fixes in the scoring script (redacted paths, network-down rows, harvest health judged from the append-only logs as of the decision time, a gate frozen once). None of them changes how τ, L, hits or confidence are computed.
6. **7 Nov 2026, 22:40 IST.** The decision time of the gate moved to Sunday 8 Nov 14:00 IST, because the harvest had run far slower than planned and τ, computed at 17:44, and L were committed only at 22:35, after the original decision time. No Test search had been run.
7. **8 Nov 2026, 17:10 IST.** The gate was not met on harvest health, and the plan said to switch to a fallback project; the author decided at about 17:08 IST to stay with Gunj, because 126 of the 150 Test streams were already harvested and no further download was needed. The blocked stream was never retried, and the gate run took place at 17:04 IST instead of 14:10, on the same streams, since the harvest had stopped at 05:30 IST.
8. **10 Nov 2026, 19:18 IST.** The bot check had come before any report-half real negative was harvested, so M3 rested on SYNTHETIC negatives alone. The author decided to retry those 25 streams once, about 62 hours after the block, signed out and with the same harvester and rules, touching no Test stream.
9. **11 Nov 2026, 18:36 IST.** The retry ran from 19:19 to 19:58 IST, met no block and harvested 24 of the 25 streams (the remaining 1 was no longer available); the frozen scoring script then recomputed M3 with the frozen τ. The libhum baseline was not run: libhum was never installed, and the time went to the harvest problems above.
10. **11 Nov 2026, 22:52 IST.** M5 on the SYNTHETIC cuts, the promised M2 and M3 plots, the M4 reading at the measured noise level and the confident-verdict breakdown come from scripts written after the Test results were known. They change no τ, L, hit, verdict or gate, and the two analyses outside the plan are labelled EXPLORATORY (post hoc). This report labels the M4 reading with v2's weighted correlation the same way, since the plan's computation uses the plain correlation.

Threats to validity. The ground truth is YouTube's release time plus L and D, since no clock inside the recordings was available. D is an estimate from a statistical clock with unknown error, measured relative to each file's first week, and L rests on 5 Calibration streams, so an offset of tens of seconds can come from any of the three or from the lock, and this evaluation cannot tell them apart. The Test streams come from institutional channels, up to 10 per channel, and may share recording conditions within a channel, while the Wilson intervals treat streams as independent. The harvest order was random, but failures may cluster by channel or stream format. The negatives are few, and each τ rests on the largest threshold-half negative of its cell. The reference is a single PMU, and v2 was tuned on 5 Dev streams.

8 Limitations and ethics

The open reference ends on 20 Aug 2023, so Gunj cannot date a later recording; Grid-India records the frequency every second, and publishing that record would allow the same check for recent recordings. Most streams carry little usable hum (M1), and a missing hum is evidence of nothing. A confident verdict can be wrong (9 of 23 in the primary cell were more than 30 s off), so it should be read with its candidate list and margin. A match dates the hum; it does not show that a recording is genuine, since anyone with the log could add a matching hum. The grid is synchronous, so the hum says nothing about where in India a recording was made, and Gunj reports time and grid only.

Court channels and proceedings were excluded, because the Gujarat High Court's live-streaming rules bar copying and derivative use of its feeds [19]; a title filter also screened for court terms in English and Hindi. The 5 Nov feasibility probe, run before this rule was written, used a lecture on that court's channel and is excluded from every split and metric. Audio was downloaded only with the author's approval, only from public institutional streams and only to measure the method, without sign-in, cookies or proxies and with a stop at the first bot check. YouTube's terms restrict downloading, so each file was deleted once its traces were extracted (the audio of the 5 Dev streams stays on the author's laptop for tuning), and no audio or video is in the repository, the explorer or this report. Dating a recording can reveal when someone attended an event, which is one reason the explorer shows only public institutional streams.

9 Conclusion

On public Indian livestreams, Gunj placed the top candidate within 30 s of the expected start for 25 of 90 excerpts of 60 min, and for 16 of 37 of those with hum. Without the correction for the log's label drift, the same endpoint was 5 of 90. Its confident verdicts were within 30 s in 14 of 23 cases, and 1 of 16 real foreign recordings reached the threshold.

Data and code availability. The code, the pre-registered protocol with its deviations, the 1-Hz traces, the raw search outputs and every result file are at https://github.com/satanrayshe/gunj, and the explorer is at https://satanrayshe.github.io/gunj/. The reference must be downloaded from Mendeley Data [2]; the SHA-256 of each file used is in `results/reference_summary.md`. No audio or video is published.

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